

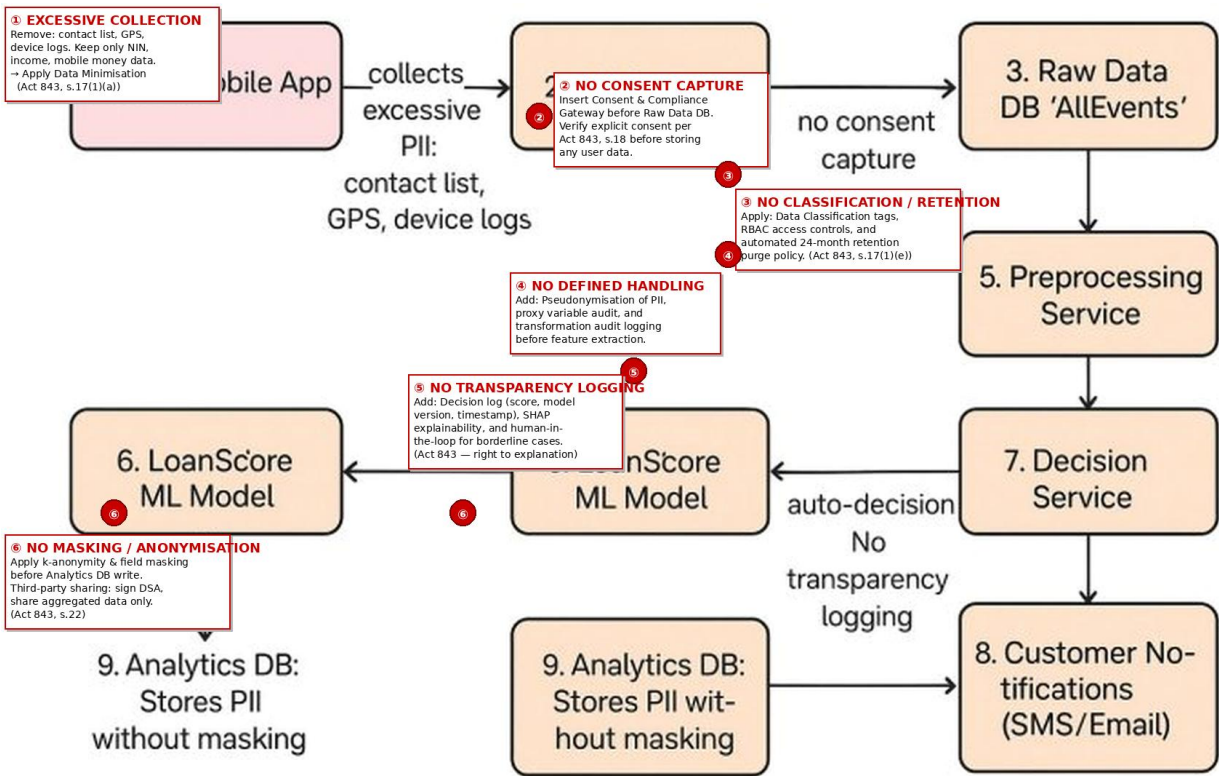
QUICKLOAN MOBILE : ETHICAL DATA GOVERNANCE REVIEW

Corrected Data Flow Diagram

Client: QuickLoan Mobile

Deliverable 2: Corrected Data Flow Diagram

The diagram below is the original QuickLoan Mobile data flow diagram with six numbered correction annotations applied directly onto the image. Each circled number (① through ⑥) marks the precise pipeline location of a governance failure. The correction details for each annotation are documented in full in the table that follows the diagram.



Annotation Key : Detailed Corrections

The table below documents the full correction for each of the six numbered annotations on the diagram above. Each entry identifies the precise pipeline step, the specific flaw present (consistent with the diagram labels), and the correction required with the Act 843 legal rationale.

#	Pipeline Step	Flaw Identified	Correction Annotation & Rationale
①	Step 1 User Mobile App (Collection Point)	Excessive PII Collection contact list, GPS location, and device logs collected beyond any loan-scoring purpose. Diagram label: "collects excessive PII: contact list, GPS, device logs."	CORRECTION : Apply Data Minimisation at Source: Remove the contact list, GPS, and device log collection permissions from the mobile app. Restrict data collection strictly to fields with a direct, documented loan-scoring purpose: (a) National ID / Ghana Card number for identity verification; (b) mobile money account details for repayment channel setup; (c) declared monthly income for creditworthiness; (d) existing loan and repayment history. WHY NECESSARY: Act 843, Section 17(1)(a) personal data must be adequate, relevant, and not excessive relative to its stated purpose. Collecting contact lists implicates thousands of third parties who have not consented to any QuickLoan data relationship, constituting unlawful processing of their personal data.
②	Step 2 → Step 3 API Gateway → Raw Data DB 'AllEvents' (Compliance Gap diagram label: "no consent capture")	No Consent Capture data is written to the Raw Data DB without any verification that the data subject has provided explicit, informed consent. This is the central legal failure of the entire pipeline.	CORRECTION : Insert Consent & Compliance Gateway: Insert a dedicated Consent & Compliance Gateway component between Step 2 (API Gateway) and Step 3 (Raw Data DB). This must: (a) verify that a valid, purpose-specific, timestamped consent record exists before any data is persisted; (b) reject and log the submission if consent is absent or expired; (c) tag every stored record with consent version, purpose scope, and expiry date; (d) honour real-time consent withdrawal. WHY NECESSARY: Act 843, Section 18 all personal data processing must be based on the explicit, informed, and freely given

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			consent of the data subject. Every record in the Raw Data DB 'AllEvents' table is currently processed without a lawful basis, rendering all downstream pipeline operations (Preprocessing, ML inference, Analytics, Third-Party Sharing) unlawful.
③	Step 3 Raw Data DB 'AllEvents' (Storage Layer)	No Data Classification, No Access Controls, No Retention Policy Sensitive/Confidential financial PII stored indefinitely with no governance controls whatsoever.	CORRECTION : Apply Classification, RBAC, and Retention Policy: Implement three controls on the Raw Data DB: (a) Data Classification tag all records with their classification label (Sensitive/Confidential for financial PII; Internal for operational metadata); (b) Role-Based Access Control (RBAC) enforce least-privilege so only the Preprocessing Service account and the authorised DPO role can read PII fields; (c) Automated Retention Enforcement purge or irreversibly anonymise raw PII records 24 months after loan closure, generating deletion certificates for audit. WHY NECESSARY: Act 843, Section 17(1)(e) personal data must be retained no longer than necessary. Without these controls the Raw Data DB is an indefinitely expanding, uncontrolled PII store representing a permanent breach surface and an unbounded compliance liability.
④	Step 5 Preprocessing Service (Feature Engineering)	No Defined Data Handling raw PII flows directly into ML feature extraction with no pseudonymisation, no proxy variable audit, and no audit	CORRECTION Pseudonymisation, Proxy Audit, and Audit Logging: Apply three controls: (a) PII Pseudonymisation replace all direct identifiers (name, NID, phone) with pseudonymous tokens

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		logging of transformations.	before feature extraction, so the LoanScore ML Model only trains on de-identified vectors; (b) Proxy Variable Audit document and review every feature; remove GPS region, contact list size, and device model as these proxy protected demographic characteristics; (c) Transformation Audit Log record every transformation operation with timestamp, input hash, and output feature vector hash for full data lineage. WHY NECESSARY: Processing raw PII in feature engineering without pseudonymisation violates Data Protection by Design principles embedded in Act 843. Proxy variables are the primary mechanism of indirect algorithmic discrimination. Audit logging is essential for Act 843 accountability obligations and for forensic investigation following any bias incident.
⑤	Step 7 Decision Service (diagram label: "auto-decision / No transparency logging")	No Transparency, No Decision Logging, No Human Oversight the LoanScore ML Model makes fully automated credit decisions with no audit trail, no explanation, and no human review at any point.	CORRECTION : Decision Logging, SHAP Explainability, Human-in-the-Loop: Implement three controls: (a) Decision Logging persist every inference record: pseudonymous applicant ID, feature vector hash, model version, credit score, decision (Approved / Referred / Rejected), and ISO 8601 timestamp; (b) Explainability (XAI) integrate SHAP values to produce a human-readable explanation of the top 3–5 features driving each decision, stored with the decision record and accessible to the applicant on request; (c) Human-in-the-Loop route all borderline decisions (score within $\pm 5\%$ of the

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			threshold) to a human credit officer for final determination. WHY NECESSARY: Fully automated credit decisions without explanation or oversight violate the right to explanation and to challenge an automated decision principles in Act 843 consistent with internationally recognised responsible AI standards. Decision logs are the foundational requirement for the DIR monitoring metric recommended in Deliverable 1.
⑥	Steps 9 & 10 Analytics DB + 3rd-Party Partner (both labelled "Stores PII without masking" in diagram)	No Data Masking or Anonymisation Sensitive/Confidential financial PII stored in the Analytics DB in identifiable form and shared with a Third-Party Partner without any de-identification, masking, or formal data sharing agreement.	CORRECTION : Masking, Anonymisation, and Data Sharing Agreement (DSA): Apply a two-stage process: (a) Analytics DB before writing, apply k-anonymity ($k \geq 5$) and field masking to all quasi-identifiers (date-of-birth → age band; exact income → income decile; GPS location → administrative region; NID → masked token); retain only aggregated, non-attributable statistical fields; (b) Third-Party Partner execute a DSA compliant with Act 843, Section 22, specifying purpose limitation, permitted data categories, retention limits, and a binding prohibition on re-identification. Share only fully anonymised, aggregated datasets never raw or pseudonymous PII. WHY NECESSARY: Storing and sharing identifiable financial PII without a lawful basis constitutes unauthorised disclosure under Act 843. Irreversible anonymisation falls outside Act 843's regulatory scope, providing a clear legal basis for analytics data sharing while

#	Pipeline Step	Flaw Identified	Correction Annotation & Rationale
			eliminating the data subject's privacy exposure.